Lecture 12 (More Simplex Method)

March 16, 2026

We will finish deriving the algorithm for the simplex method today. Next class we will talk about how to deal with degenerate solutions.

Notation

Suppose we have an LP of the form:

$$\text{minimize } \vec{c} \cdot \vec{x}$$

Subject To

$$A\vec{x} = \vec{b}$$

$$\vec{x} \geq 0$$

If we have a BFS to this system, we can take $B(1), B(2), \ldots, B(m)$ to be the indices of the m active variables. This yields the matrix

$$B = \begin{pmatrix} A_{B(1)} & A_{B(2)} & \ldots & A_{B(m)} \end{pmatrix}$$

Where each A_i is the ith column of A. We also have the reduced cost

$$\bar{c}_j = c_j - \vec{c}_B \cdot \left(B^{-1} A_j\right)$$

where $\vec{c}_B$ is the vector $\vec{c}_B = \begin{pmatrix} c_{B(1)} & c_{B(2)} & \ldots & c_{B(m)} \end{pmatrix}$. For convenience, we will call the corresponding feasible region, P. We say a basis matrix, B, is optimal if B satisfies:
a) $B^{-1}\vec{v} \geq 0$
b) $\vec{\bar{c}} = \vec{c} - \vec{c}_B^T \left(B^{-1} A\right) \geq 0$

More Derivation

For today's lecture, we will again assume every BFS is non-degenerate. We will deal with the cases where this isn't true later. We will also assume that P is non-empty.

Case 1)

If we have a BFS, $\vec{x}$, and $\bar{c}_j \geq 0$ for all j at $\vec{x}$, then $\vec{x}$ is optimal and we are done.

Case 2)

If we have a a BFS, $\vec{x}$, and for some j we see $\bar{c}_j < 0$, then we have a feasible direction. This direction is $\vec{d}$ (which we defined last lecture). This is the vector with

$$d_j = 1,\ d_i = 0 \text{ if } i \neq B(1), B(2), \ldots, B(m)$$

and with

$$\vec{d}_B = -B^{-1} A_j$$

If we move along this direction, then we see $\vec{x}_B + \theta\vec{d}_B \geq 0$ (as long as we take a small enough step, θ), $x_i + \theta d_i = 0$ for all $i \neq j, B(1), \ldots B(m)$ (As $x_i = d_i = 0$ for these indices), and $x_j b + \theta d_j > 0$ (as $d_j = 1$ and $x_j = 0$). When we move along this direction as far as we can while remaining in P, then we say x_j (or A_j) has entered the basis. If we move as far as possible, we move to the new BFS,

$$\vec{x} + \theta^* \vec{d}$$

where

$$\theta^* = \max\left\{\theta \geq 0 | \vec{x} + \theta\vec{d} \in P\right\}$$

We do have a few nice properties around θ. Notice

$$\vec{c} \cdot \theta^* \vec{d} = \theta^* \left(\vec{c} \cdot \vec{d}\right) = \theta^* c_j$$

How can we find θ^* efficiently? Consider

$$A\left(\vec{x} + \theta\vec{d}\right) = A\vec{x} + \theta A\vec{d} = A\vec{x} = \vec{b}$$

as we assumed $A\vec{d} = 0$ in the derivation of $\vec{d}$ last lecture. This means we only need to check when $\vec{x} + \theta\vec{d}$ gets negative components to check for feasibility. We have two cases:

Case 1) If $\vec{d} \geq 0$, then $\vec{x} + \theta\vec{d} \geq 0$ for all $\theta \geq 0$. This means that $\vec{x} + \theta\vec{d}$ is always feasible and we can keep moving in the direction, $\vec{d}$ indefinitely. This implies that P contains a non-trivial cone and that $\theta^* = \infty$.

Case 2) If there exists i such that $d_i < 0$, then the restriction $x_i + \theta d_i \geq 0$ implies

$$\theta \leq -\frac{x_i}{d_i}$$

So

$$\theta^* = \min_{\{i | d_i < 0\}} \left(-\frac{x_i}{d_i}\right)$$

We can make this formulation tighter though. We know $d_j = 1$ and $d_i = 0$ if $i \neq B(1), B(2), \ldots, B(m)$. As such, we only need to check some of the indices.

$$\theta^* = \min_{\{i=1,\ldots,m | d_{B(i)}\}} \left(-\frac{x_i}{b_i}\right)$$

In both cases, we see $\theta^* > 0$. Now that we know θ^*, we can update the active variables. Let l be the minimizing index (i.e.)

$$-\frac{x_{B(l)}}{d_{B(l)}} = \min_{\{i=1,\ldots,m | d_{B(i)} < 0\}} \left(-\frac{x_{B(i)}}{d_{B(i)}}\right) = \theta^*$$

Let $\vec{y} = \vec{x} + \theta^* \vec{d}$. Then since $d_{B(l)} < 0$ and $y_l = x_{B(l)} + \theta^* d_{B(l)} = 0$ we can get a new basis matrix

$$\bar{B} = \begin{pmatrix} A_{B(1)} & \ldots & A_{B(l-1)} & A_j & A_{B(l+1)} & \ldots & A_{B(m)} \end{pmatrix}$$

by replacing the lth column of B with the jth column of A. This also gives us new basis indices:

$$\bar{B}(i) = \begin{cases} B(i) & i \neq j \\ j & \text{if } i = l \end{cases}$$

Theorem

1. The columns of $\bar{B}$ are linearly independent.
2. $\vec{y} = \vec{x} + \theta^* \vec{d}$ is a BFS associated with $\bar{B}$.

Proof

1. If $A_{\bar{B}(i)}$ for $i = 1, \ldots, m$ are linearly independent, then we know

$$\sum_{i=1}^{m} \lambda_i A_{\bar{B}(i)} = 0 \iff \lambda_i = 0 \ \forall i$$

As B is invertible, this is equivalent to stating

$$\sum_{i=1}^{m} \lambda_i B^{-1} A_{\bar{B}(i)} = 0 \iff \lambda_i = 0 \ \forall i$$

So if we instead prove that $B^{-1}A_{\bar{B}(i)}$ for $i = 1, \ldots, m$ are linearly independent, then we will be able to prove the original claim. As B is invertible, we know

$$B^{-1}B = I$$

Thus

$$B^{-1}A_{\bar{B}(i)} = B^{-1}B_i = \vec{e}_i$$

when $i \neq l$. Here, we have $\vec{e}_i$ to be the ith unit vector. For $i = l$ we instead have

$$B^{-1}A_{\bar{B}(l)} = B^{-1}A_j = -\vec{d}_B$$

But as $y_l = 0$, we know $-d_{B(l)} \neq 0$. As each unit vector $\vec{e}_i$ is linearly independent to the other unit vectors and each of these $\vec{e}_i$ contains 0 for the lth entry and $-\vec{d}_B$ doesn't, then we know that we have a linearly independent set of vectors. Thus $\bar{B}$ has a linearly independent set of columns, thus $\bar{B}$ is invertible and a basis matrix.

2. As $\vec{y} \geq 0$ and $A\vec{y} = \vec{b}$ and $y_i = 0$ for all $i \neq \bar{B}(1), \ldots, \bar{B}(m)$ and since $A_{\bar{B}(1)}, \ldots, A_{\bar{B}(m)}$ is linearly independent, then $\vec{y}$ is a BFS associated with $\bar{B}$. (Note that $A_{\bar{B}(1)}, \ldots, A_{\bar{B}(m)}$ is the set of columns of $\bar{B}$).

The Simplex Method (Finally)

For convenience we will call $\vec{u} = -\vec{d}_B = B^{-1}A_j$. Here is the process for one iteration (**pivot**) of the simplex method.

1) Start with a basic solution $\vec{x}$ with a basis $A_{B(1)}, \ldots, A_{B(m)}$.

2) Compute the reduced cost $\bar{c}_j = c_j - \vec{c}_B^T B^{-1} A_j$ for all non-basic indices, j. If $\bar{c}_j \geq 0$ for all j, then $\vec{x}$ is optimal and we terminate the algorithm. Otherwise, select one j such that $\bar{c}_j < 0$. We will refer to j as this selected j for the remainder of the algorithm.

3) Compute $\vec{u} = B^{-1}A_j$. If no component of $\vec{u}$ is positive, we have $\theta^* = \infty$, thus $-\infty$ is the optimal solution (i.e. the LP is unbounded). If this is the case, terminate the algorithm.

4) If some component of $\vec{u}$ is positive, then get

$$\theta^* = \min_{\{i=1,\ldots,m | u_i > 0\}} \frac{x_{B(i)}}{u_i}$$

5) Let l be such that $\theta^* = \frac{x_{B(l)}}{u_l}$. Form the new basis by replacing $B_{(l)}$ with j and get the new basis matrix by replacing the lth column of B with the jth column of A. Also update $\vec{x}$ by taking $\vec{x} + \theta^* \vec{d} \to \vec{x}$.

Notes

As each pivot will strictly decrease the objective, then each pivot cannot visit a previously visited BFS. As any polyhedron has finitely many vertices, then there are only finitely many BFS for thee simplex method to visit. Eventually the simplex method will reach an optimal BFS or the simplex method will show the LP is unbounded.